Spondyloarthritis
Nov 21, 2025
Classification Criteria, Features, Therapies

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Case 1

CC: "my back hurts"

HPI: 25 yo male with 4 mos of bilateral buttock and lower back pain. Dull, achy but can be sharp with coughing, sneezing, bending; radiates to mid thigh; 4 hours am stiffness. Pain improves with hot showers, walking.

ROS: No ankle pain, eye redness, STD, rash, bloody diarrhea

SH: No tob/etoh/drugs

Labs: CBC, CMP normal, ESR: 40 mm/hr (<30 mm/hr)

Plain films of L spine and pelvis: normal


BPR
L
HO, TONY DONGSUN
Plate ID
15 CEPHA


AGE: 027Y
SEX: M


Plate ID: 9102145783

What is the most likely cause of this young man’s symptoms?

1. Spondylolisthesis
2. Spondylosis
3. Lumbago
4. Inflammatory back pain
5. Fibromyalgia

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What features make his back pain inflammatory?

1. His youth
2. Male sex
3. Morning stiffness
4. Improvement with exercise
5. Normal plain films

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Common causes of back pain

- Lumbar strain
- Degeneration of spine
- Herniated disc
- Spinal stenosis
- Spondylolisthesis
- Osteoporotic compression fractures
- Spondylosis

Image courtesy of Dr. Marina Magrey.

1. Deyo JW, Weinstein JN. *N Engl J Med.* 2001;344:363-370.

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Mechanical vs inflammatory back pain

Mechanical back pain (MBP)	Inflammatory back pain (IBP)
▪ Pain arising intrinsically from the spinal joints, intervertebral disks, or soft tissues[1] ▪ May be chronic but is usually acute and self limiting[2]	▪ Due to an underlying disease indicative of inflammation of the vertebrae, joints of the spine, and entheses[2] ▪ Age of onset is typically < 40 years; results in chronic back pain lasting > 3 months[2]

1. Will JS, et al. *Am Fam Physician.* 2018;98:421-428. 2. Sieper J, et al. *Ann Rheum Dis.* 2009;68:784-788.

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Case 2

CC: “my gout is acting up for the past couple weeks”

HPI:

- 64 yo man previously diagnosed “gout”: intermittent swelling of various toes and achilles tendon
- alternating feet, ankles x 20 years for weeks at a time
- ibuprofen helps usually
- “I religiously take my medicines and monitor my diet”

Exam and Labs

MSK:

- Moderate swelling in R fifth digit in foot
- Osteoarthritic changes in fingers

SKIN:

- No psoriasis

Labs:

- CBC, BUN/CRE normal
- ESR 15 mm/hr, CRP: 25.0 mg/L
- Uric acid 5.2

What is the diagnosis?

1. Calcium pyrophosphate arthritis (CPPD)
2. Gout
3. Rheumatoid Arthritis
4. Spondyloarthritis
5. Osteoarthritis

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Learning Objectives

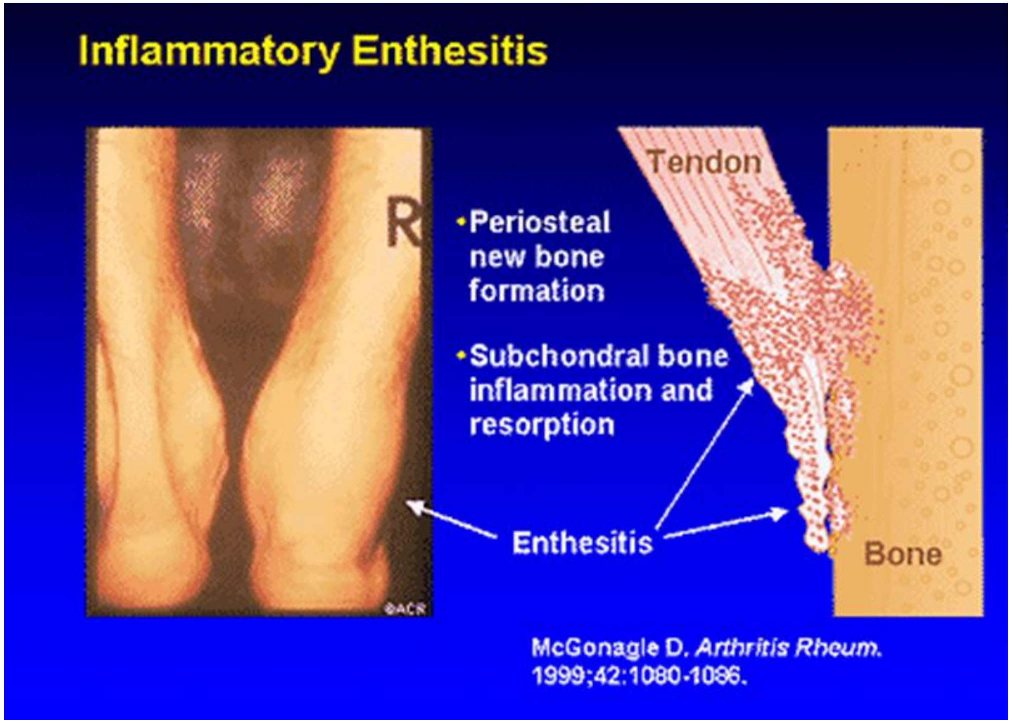
Spondyloarthritis (SpA)

- Epidemiology
- Classification:
    - Axial disease (radiographic and nr-SpA)
    - Peripheral disease
- Early Diagnosis
- Pathogenesis
- Radiographs
- Treatments

SpA: General Points

Highly heterogeneous presentation

- Inflammatory low back pain
- Sacroileitis
- Dactylitis
- Enthesitis



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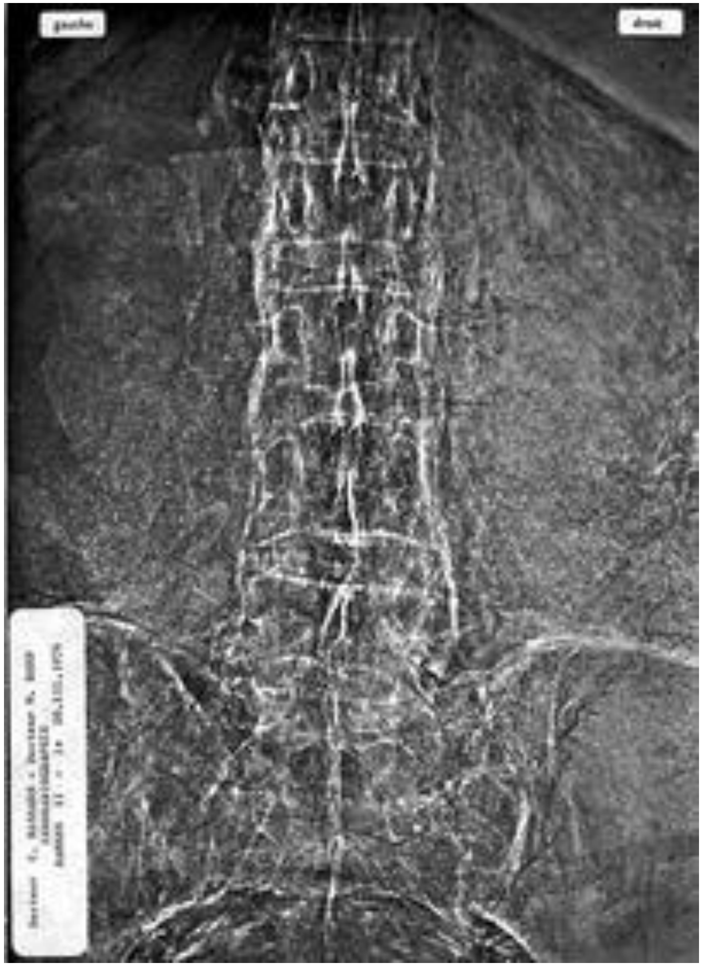
Historical Perspective



Ankylosing spondylitis in the pharaohs of ancient Egypt. Feldtkeller E. Rheumatol Int. 2003

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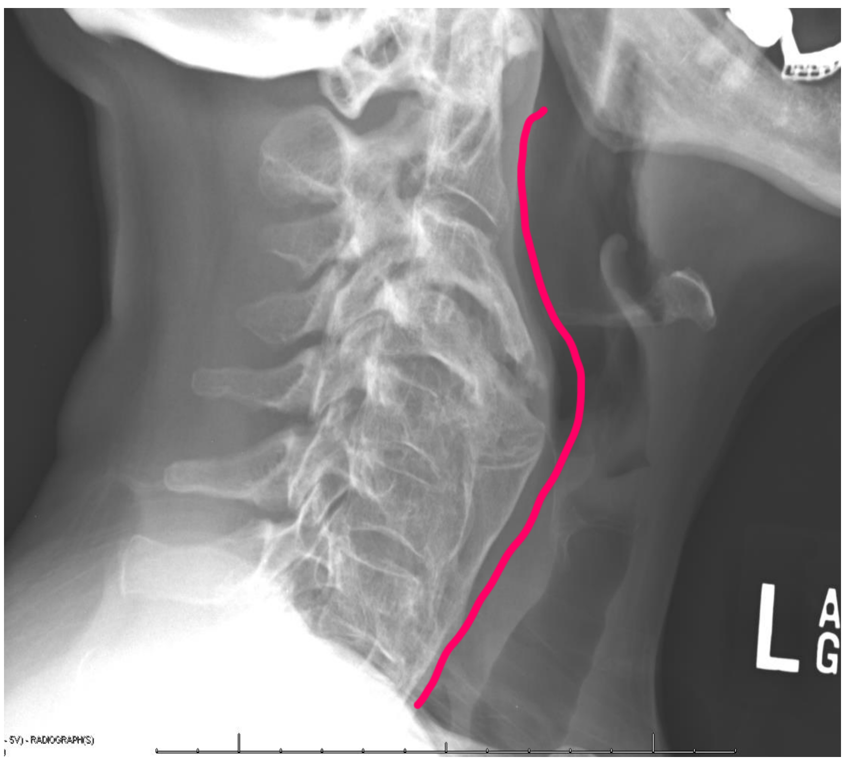
3rd case

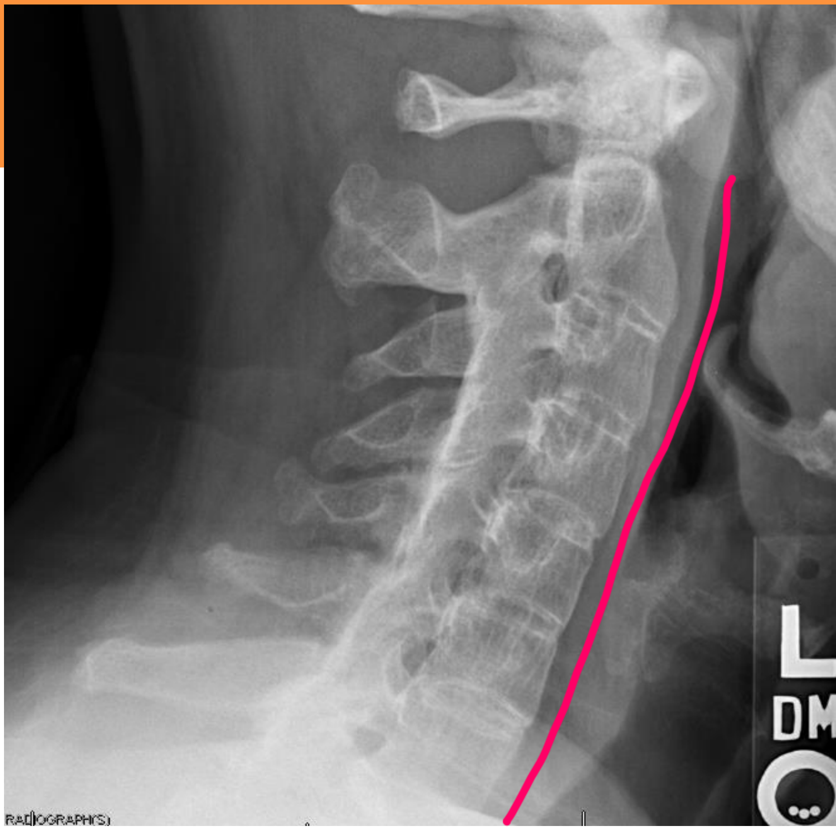
64 yo man with longstanding DM presents with 6 months of worsening chronic neck stiffness

Diffuse Idiopathic Skeletal Hyperostosis (DISH)

"Candle wax pattern":

1. presence of continuous ossification/calcification along anterolateral aspects of 4 contiguous vertebrae
2. preservation of disks without degeneration
3. absence of facet joint fusion or sacroiliac erosions/fusion





Ankylosing Spondylitis

- Marginal syndesmophytes

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Diffuse Idiopathic Skeletal Hyperostosis (DISH)

- Commonly mistaken for Spondyloarthritis
- Age of onset >50
- Genetic associations
- Endocrinopathies (DMII, PTH)
- Ossification of ALL, extraspinal ligaments
- No role for immunomodulators
- Analgesia, NSAIDs, PT, surgical options PRN

Chakravarty SD, Rheumatic manifestations of endocrine disease. Curr Opin Rheumatol. 2013

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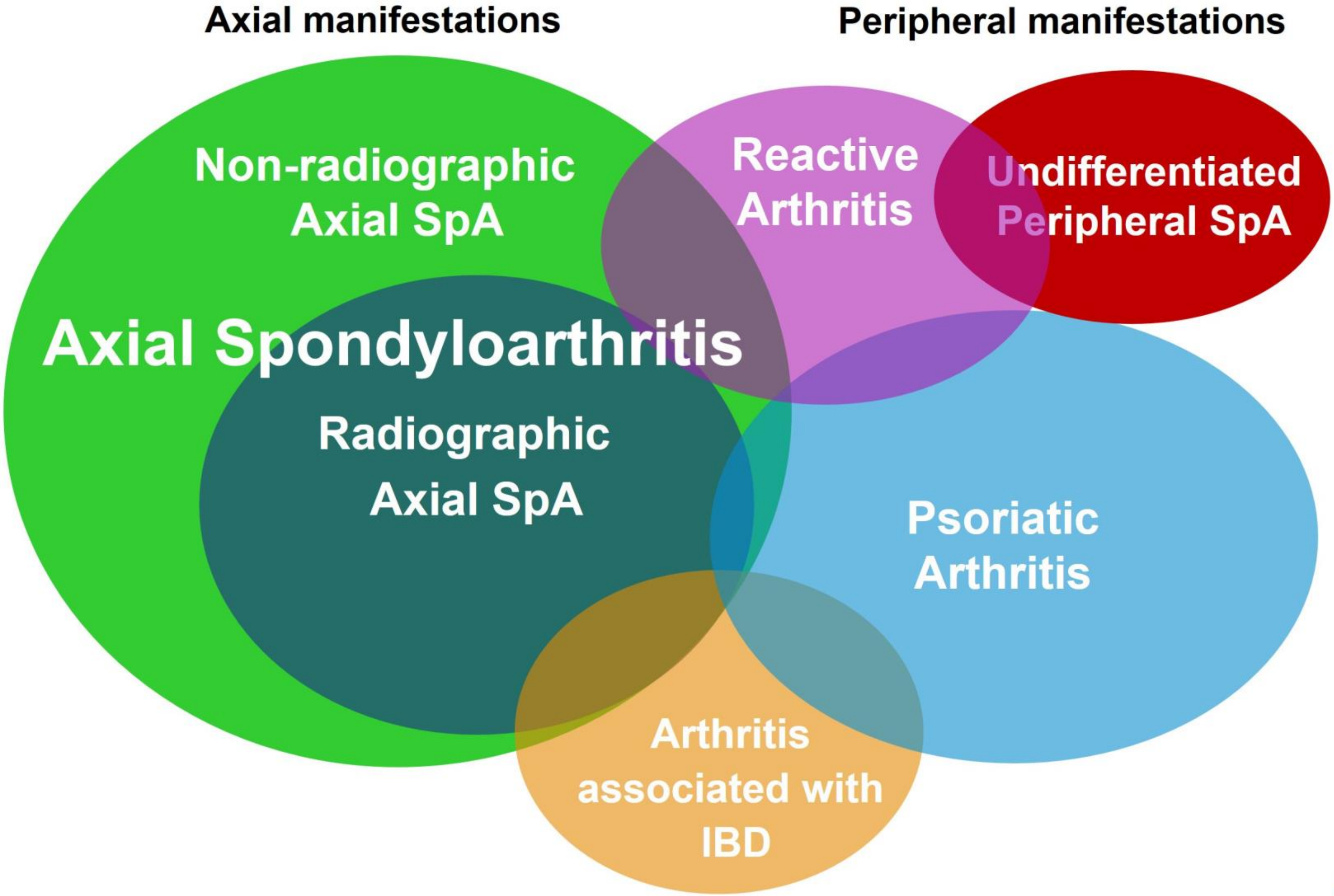
Spondyloarthritides (SpA)



Modified from Proft F et al. Ther Adv Musculoskelet Dis 2018;10:129-39

Concept of Spondyloarthritides (SpA)

Non-radiographic
axial SpA

Ankylosing Spondylitis

Reactive arthritis

Psoriatic Arthritis

Arthritis with inflammatory
bowel disease

Undifferentiated SpA

Predominantly Axial
SpA

Predominantly Peripheral
SpA

ASAS Inflammatory Back Pain Criteria by Experts (Chronic Back Pain; n=648)

- age at onset < 40 years
- insidious onset
- improvement with exercise
- no improvement with rest
- pain at night (with improvement upon getting up)

Sensitivity: 79.6%; Specificity: 72.4%

Inflammatory back pain present if <u>at least 4 out of 5</u> parameters are fulfilled.

Sieper J et al. Ann Rheum Dis. 2009;68:784-788

Spondylitic Disease Without Radiographic Evidence of Sacroiliitis. 1985

Khan MA, et al. Arthritis Rheum. 1985; 28: 40-3.

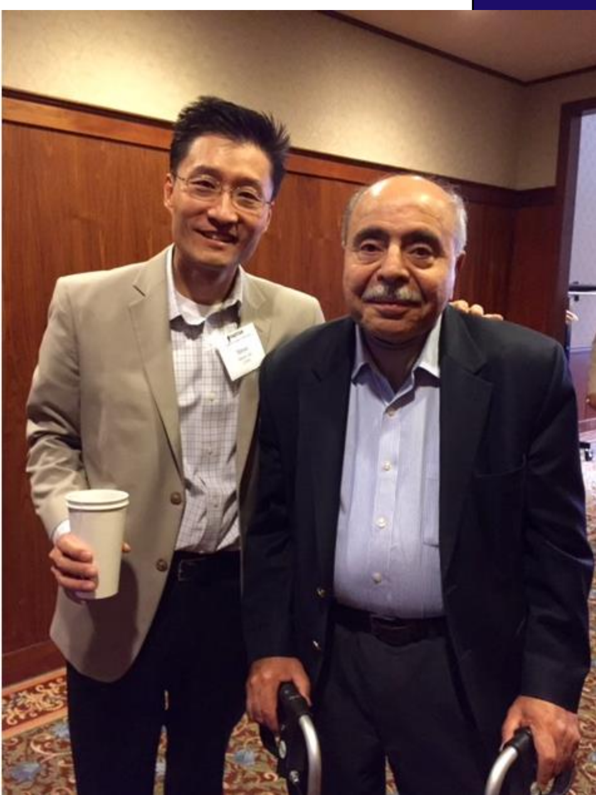
Ankylosing Spondylitis (Modified New York Criteria). 1984

Van Der Linden S, et al. Arthritis Rheum. 1984; 27:361-8

Chronic inflammatory back pain (IBP)

IBP
Radiographic sacroiliitis

Syndesmophytes



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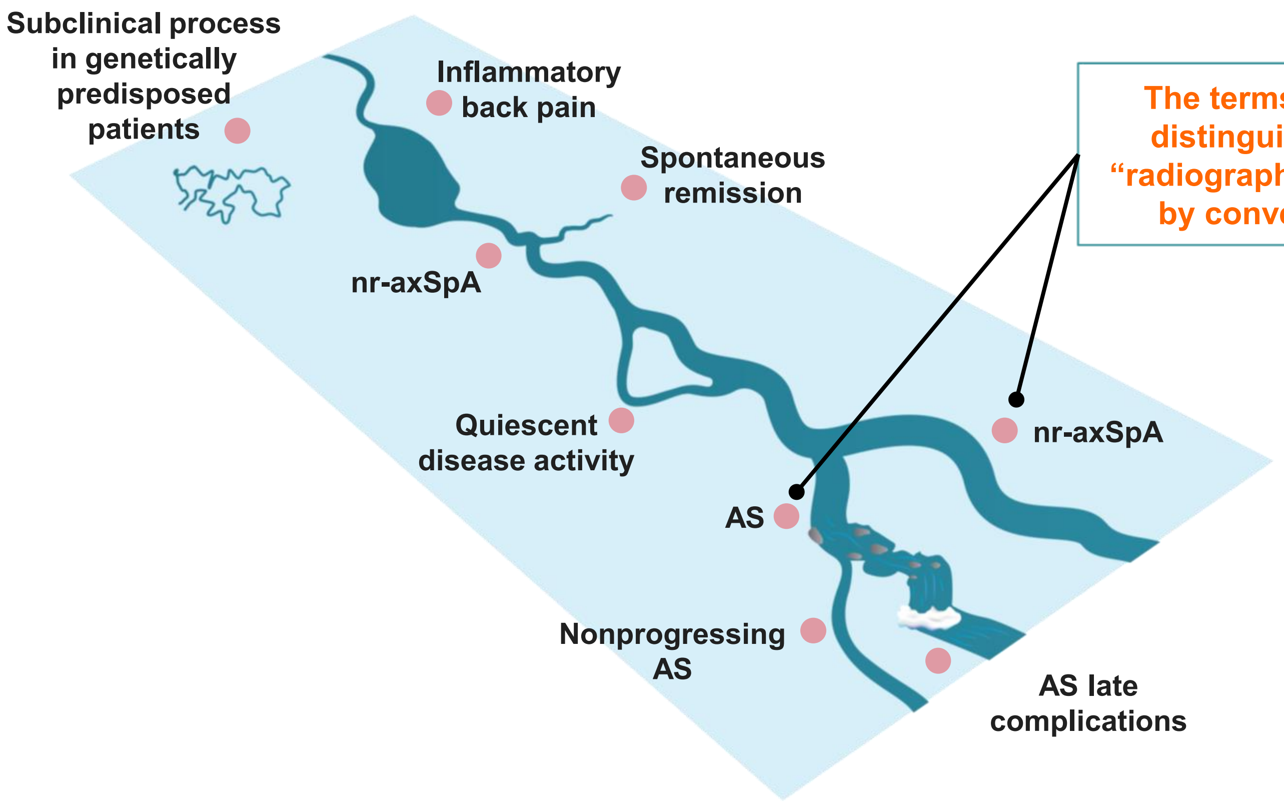
The natural history of axSpA includes AS and nr-axSpA[1]



The terms AS and nr-axSpA are distinguished by the degree of "radiographic sacroiliitis" assessed by conventional radiography[2]

Figure adapted with permission from Garg N, et al.[1]

1. Garg N, et al. *Best Pract Res Clin Rheumatol*. 2014;28:663-672. 2. Deodhar A, et al. *Ann Rheum Dis*. 2016;75:791-794.

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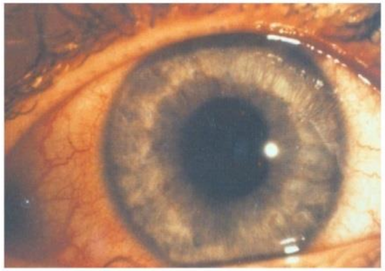
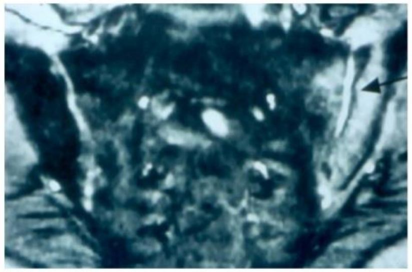
Spondyloarthritis: Characteristic Parameters Used for Diagnosis I

Symptoms

Inflammatory back pain



Imaging



Lab

ESR/CRP

Patient‘s history

Good response to NSAIDs

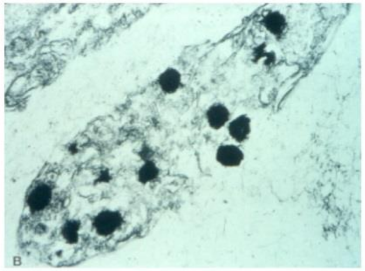
Spondyloarthritis: Characteristic Parameters Used for Diagnosis II

Genetics

HLA-B27 positive

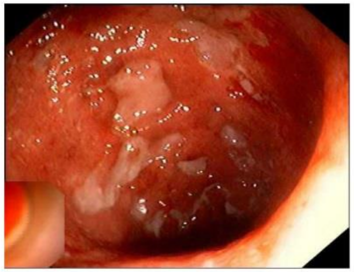
family history

Predisposing/ concomitant diseases



Infection*

psoriasis



Crohn‘s

*positive staining for Chlamydia in synovial membrane[1]

1. Schumacher HR et al. Arthritis Rheum 1988;31:937-946

Keratoderma Blenorrhagicum

ASAS Classification Criteria for Spondyloarthritis (SpA)

In patients with ≥3 months back pain and age at onset <45 years

Sacroiliitis on imaging plus ≥1 SpA feature

OR

HLA-B27 plus ≥2 other SpA features

SpA features
- inflammatory back pain (IBP)
- arthritis
- enthesitis (heel)
- uveitis
- dactylitis
- psoriasis
- Crohn's/colitis
- good response to NSAIDs
- family history for SpA
- HLA-B27
- elevated CRP

Sensitivity: 79.5%, Specificity: 83.3%; n=975

In patients with peripheral symptoms ONLY

Arthritis or enthesitis or dactylitis

plus

≥1 SpA feature
- uveitis
- psoriasis
- Crohn's/colitis
- preceding infection
- HLA-B27
- sacroiliitis on imaging

OR

≥2 other SpA features
- arthritis
- enthesitis
- dactylitis
- IBP ever
- family history for SpA

Rudwaleit M et al. Ann Rheum Dis 2011;70:25-31 (with permission)

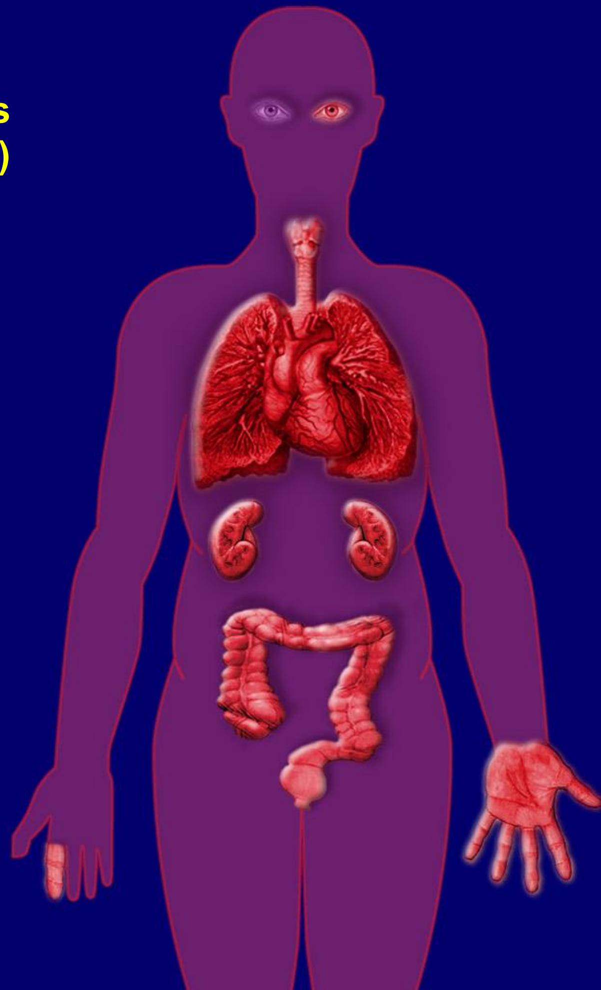
Potential Other Disease Manifestations in SpA

Eyes
(Acute Anterior Uveitis)

Heart
(Aortic Insufficiency, Heart Block)

Gut
(Inflammatory Bowel Disease,
Microscopic Inflammatory Lesion)

Dactylitis



Lungs
(Restrictive Lung Disease,
Apical Fibrocystic Disease)

Kidneys
(Amyloidosis)

Skin
(Psoriasis & Nail Changes)

Cauda Equina Syndrome

Osteopenia

HLA-B27 disease associations

Ankylosing spondylitis	**> 90% (white males)**
▪ with uveitis or aortitis	~100%
Reactive arthritis	50-80%
▪ with sacroiliitis or uveitis	90%
Juvenile spondylarthropathy	80%
Inflammatory bowel disease	
▪ Peripheral	Not increased
▪ Axial	
▪ Crohn's disease	50%
▪ Ulcerative colitis	70%
Psoriatic	
▪ Peripheral	Not increased
▪ Axial	50%

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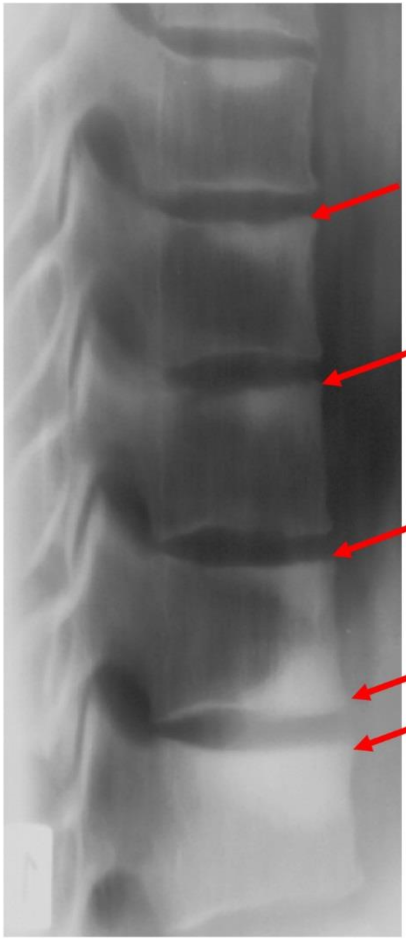
Evidence of Chronic Spinal Changes in Ankylosing Spondylitis

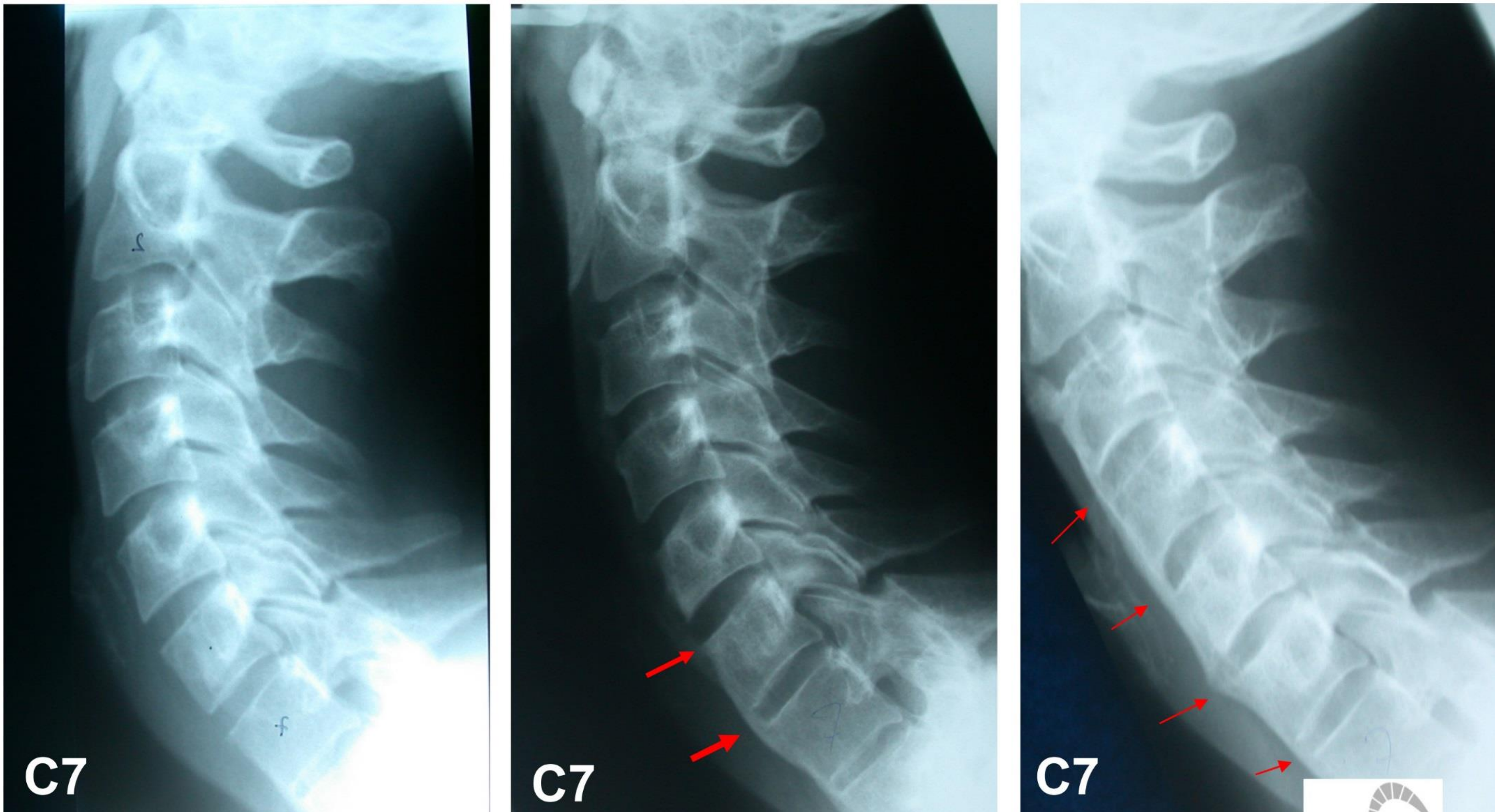


Sclerosis
„shiny corners“

Syndesmophytes
(and spondylophytes)

Bridging
syndesmophytes

Progression of Cervical Syndesmophytes 2-Year Intervals



AS, m, 30 y, disease duration 14 y

ASAS handbook, Ann Rheum Dis 2009; 68 (Suppl II) (with permission)

Atypical patient of mine

2016

48

2020

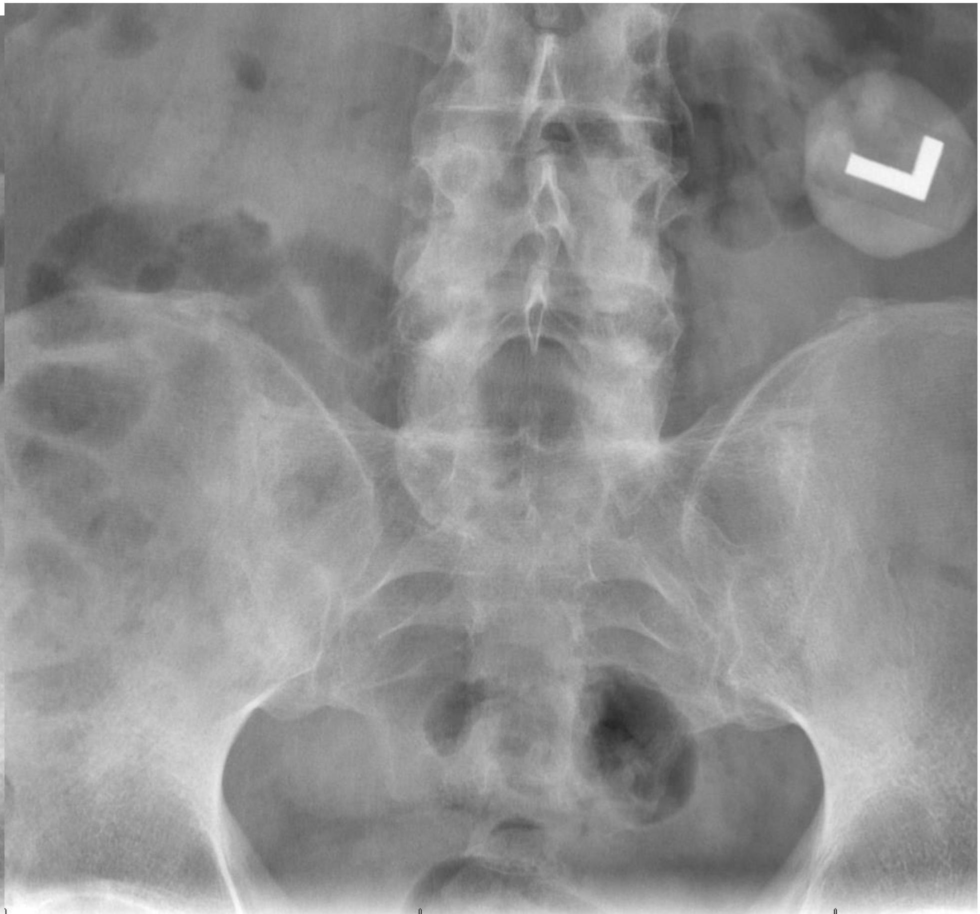
52

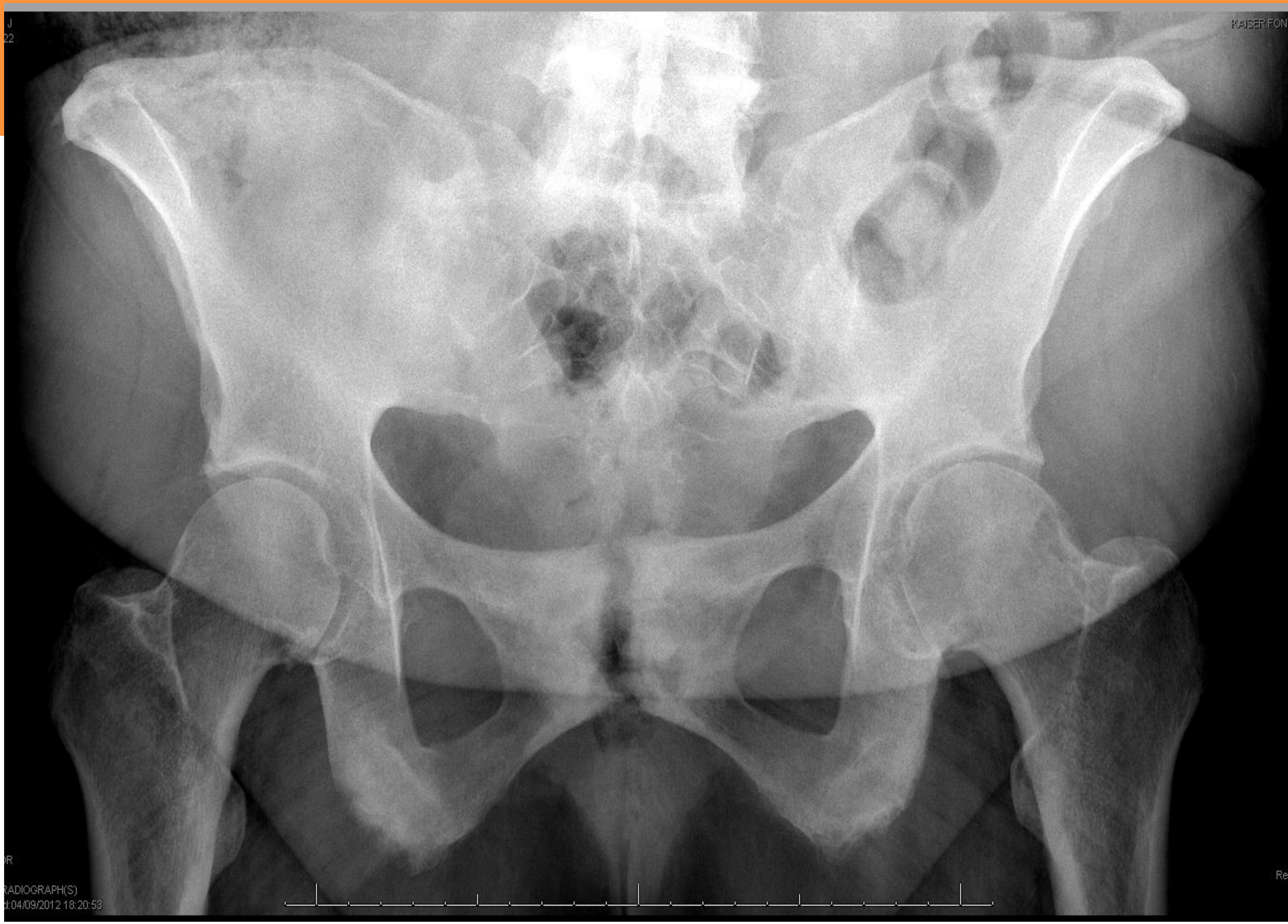
2024

56

2016

2020





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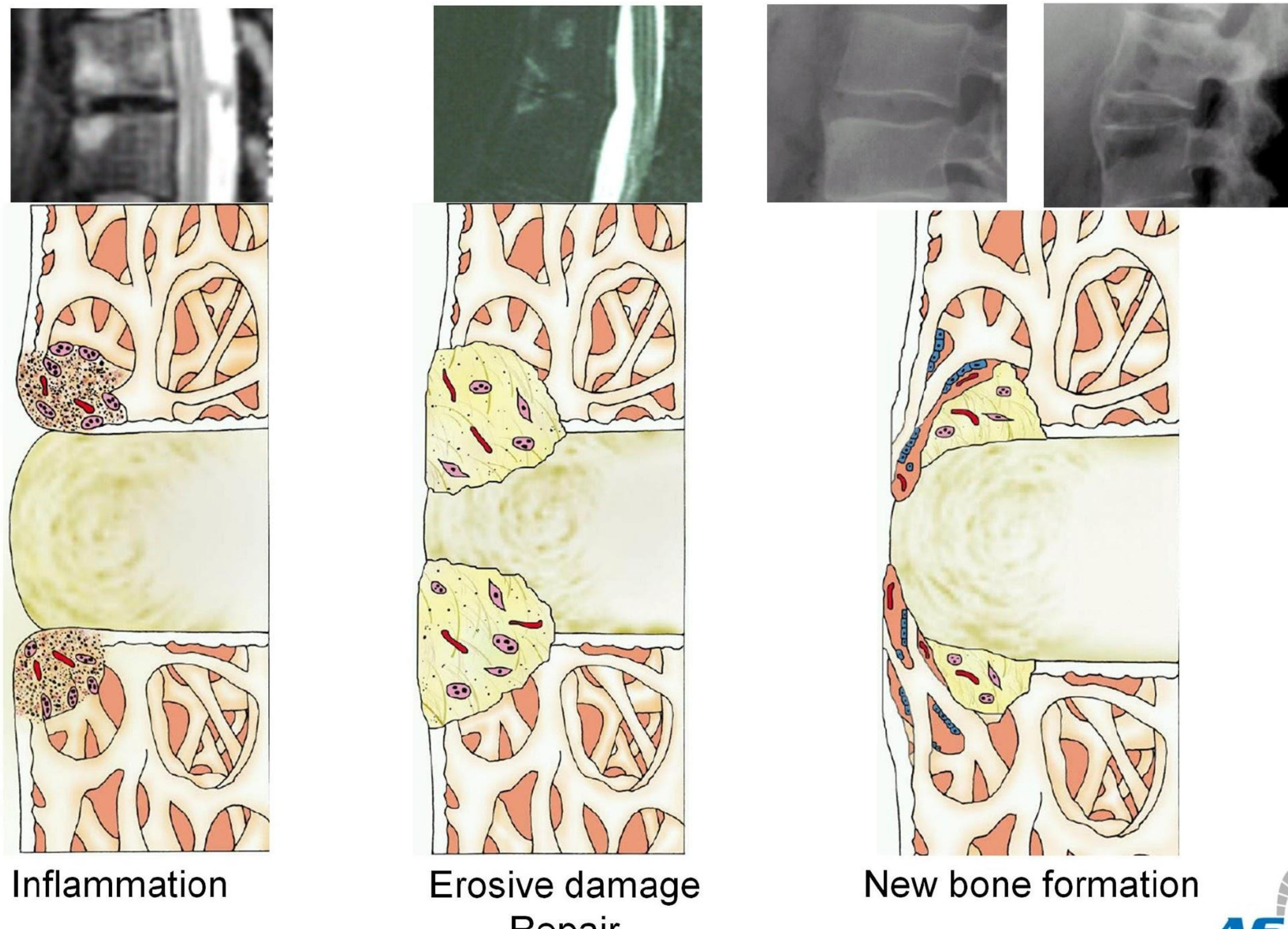
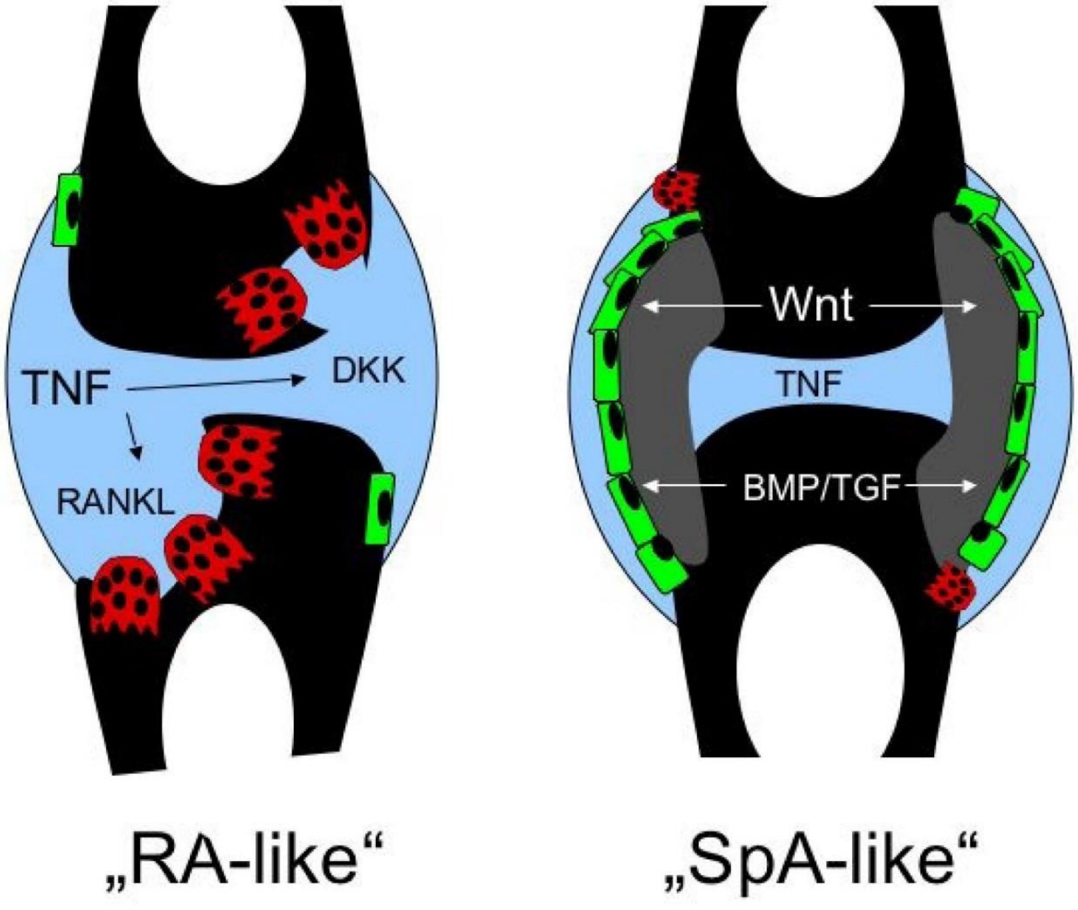
Proposed Sequence of Structural Damage in Ankylosing Spondylitis



Appel H & Sieper J et al. Curr Rheumatol Rep. 2008;10:356-63 (with permission)

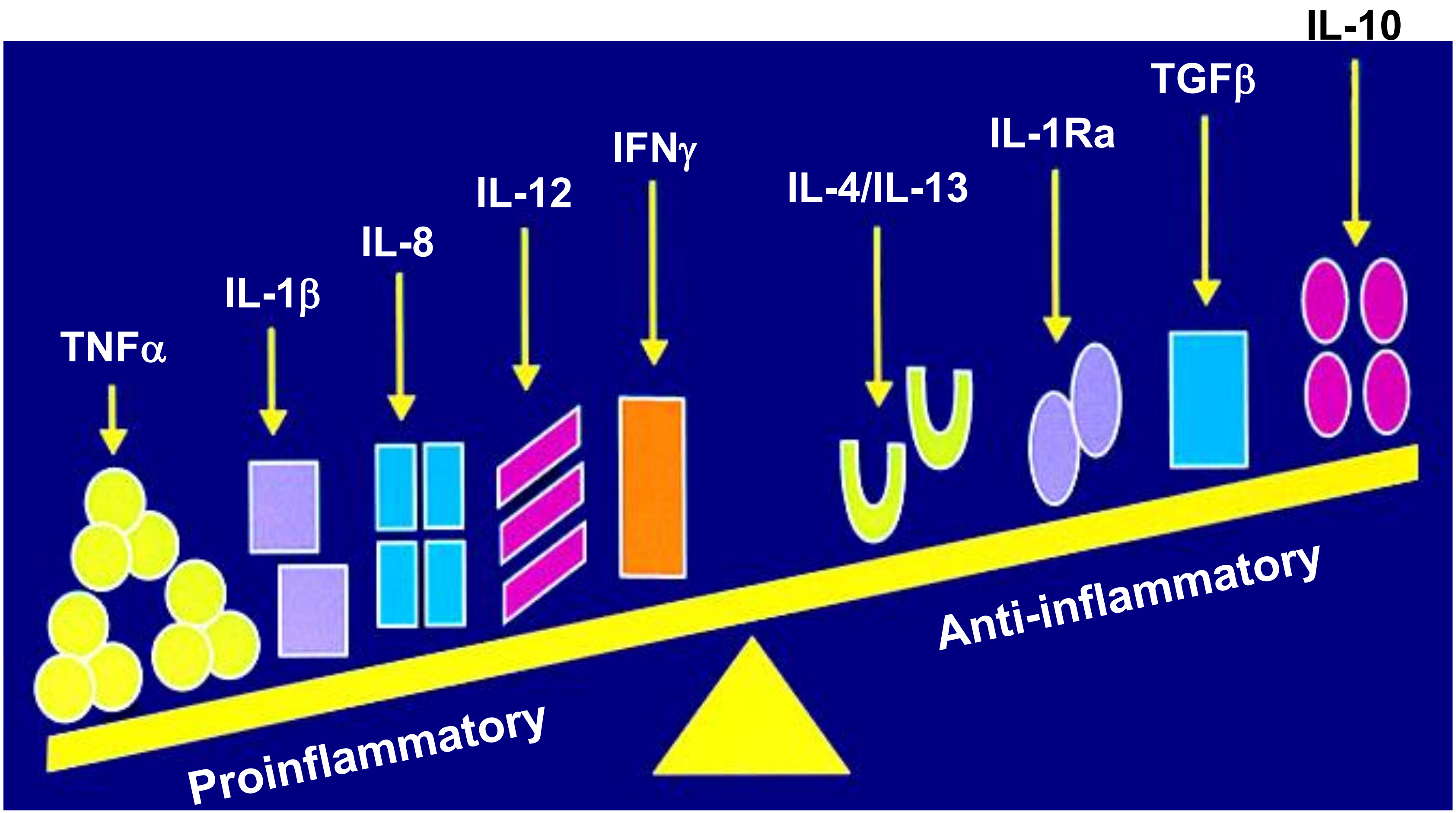
Inflammation and New Bone Formation:

Differences Between Rheumatoid Arthritis (RA) and Spondyloarthritis (SpA)



Schett G et al. Ann Rheum Dis 2007;66:709-11 (with permission)

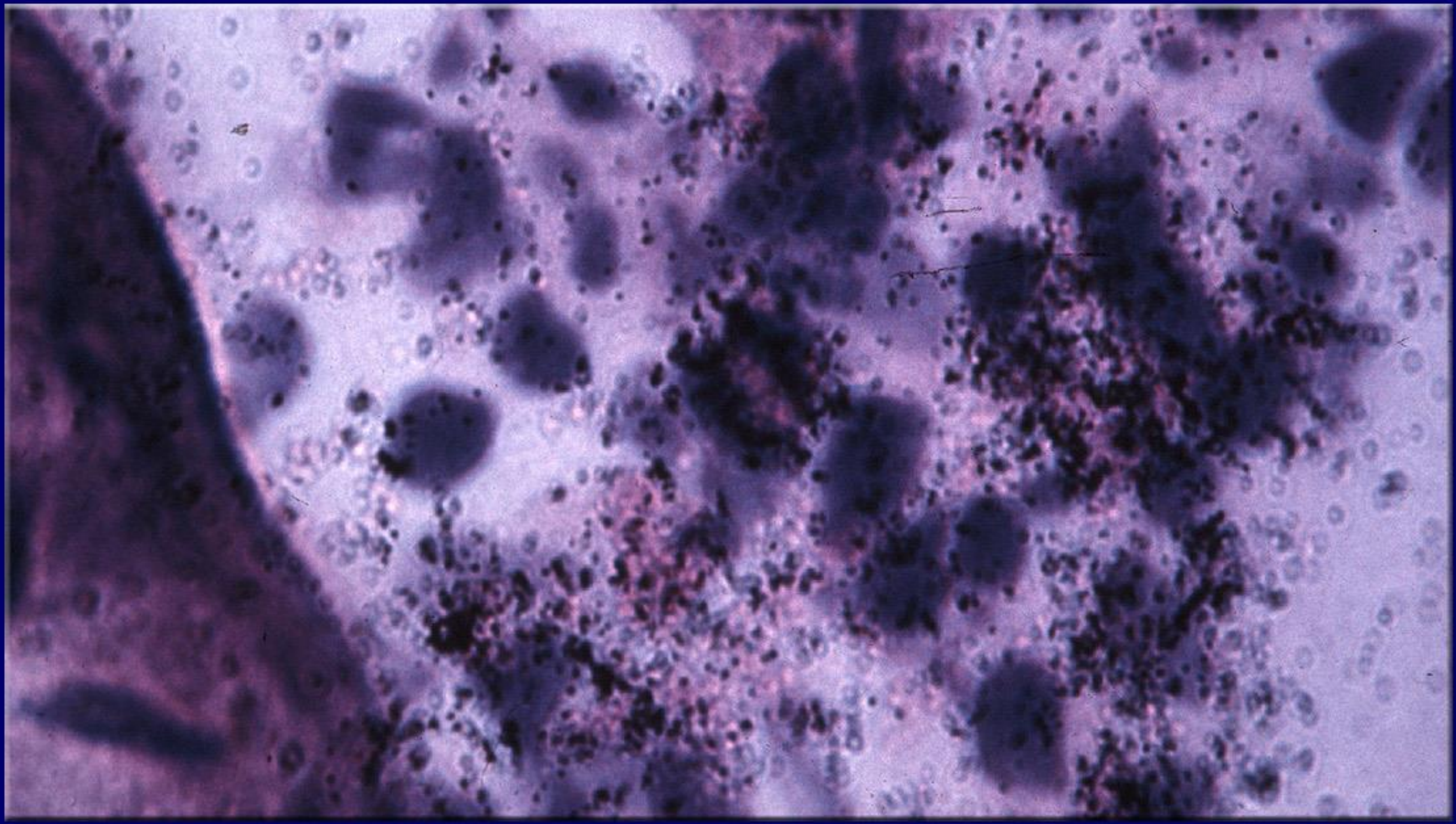
Chronic Inflammation: Imbalance of Mediators



Mcinnes IB, Schett G. *Nature Reviews*. 2007;7:429-442.
Tracey D, et al. *Pharmacology & Therapeutics*. 2008;117:244-279.

Role of TNF-α in AS

TNF-α mRNA in Sacroiliac Biopsy



Reprinted with Permission from Braun J. et al, *Arthritis and Rheumatism* 1995;38:499-505.

ASAS-EULAR Recommendations for the Management of Axial Spondyloarthritis

Axial disease

Peripheral disease

Education, exercise, physical therapy, rehabilitation, patient associations, self help groups

NSAIDs

Sulfasalazine

Local corticosteroids

bDMARDs (TNFα and IL-17 inhibitors)

Analgesics

Surgery

Jak's: tofacitinib, upadacitinib

van der Heijde D et al. Ann Rheum Dis 2017;76:978-991 (with permission)

SPECIAL ARTICLE

2019 Update of the American College of Rheumatology/Spondylitis Association of America/Spondyloarthritis Research and Treatment Network Recommendations for the Treatment of Ankylosing Spondylitis and Nonradiographic Axial Spondyloarthritis

Michael M. Ward,[1] Atul Deodhar,[2] Lianne S. Gensler,[3] Maureen Dubreuil,[4] David Yu,[5] Muhammad Asim Khan,[6] Nigil Haroon,[7] David Borenstein,[8] Runsheng Wang,[9] Ann Biehl,[1] Meika A. Fang,[10] Grant Louie,[11] Vikas Majithia,[12] Bernard Ng,[13] Rosemary Bigham,[14] Michael Pianin,[15] Amit Aakash Shah,[16] Nancy Sullivan,[17] Marat Turgunbaev,[16] Jeff Oristaglio,[17] Amy Turner,[16] Walter P. Maksymowych,[18] and Liron Caplan[19]

Arthritis Care & Research
Vol. 71, No. 10, October 2019, pp 1285–1299
DOI 10.1002/acr.24025

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Empowering Rheumatology Professionals

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SpA: Biologic DMARD's

TNFα antagonists:

- Etanercept (Enbrel)
- Infliximab (Remicade)
- Adalimumab (Humira)
- Golimumab (Simponi)
- Certolizumab (Cimzia)

IL-23 inhibitor (PsO, PsA)

- Guselkumab (Tremfya)
- Risankizumab (Skyrizi)

T cell costimulation (PsA)

- Abatacept (Orencia)

•IL-17 inhibitors (PsO, PsA, SpA)

- Secukinumab (Cosentyx)(2015)
- Ixekizumab (Taltz)
- Brodalumab (Siliq)
- Bimekizumab (Bimzelx) (sept 2024)

•IL-12, 23 inhibitors (PsO and PsA)

- Ustekinumab (Stelara)

•PDE inhibitor (PsO and PsA)

- Apremilast (Otezla)

•Biosimilars and JAK's…..

Safety and Biologics

Serious Infections

Opportunistic infections (TB, cocci, histo, blasto)

Malignancy/lymphoma

Demyelinating syndrome

Hematologic abnormalities

Congestive heart failure

Hepatic

IBD (il-17)

Auto-antibodies and drug induced lupus

Neutralizing antibodies

Spondyloarthritis Summary….

Highly heterogeneous genetic family of conditions

Recognize key clinical features

- Inflammatory vs mechanical back pain
- IBD, uveitis
- skin, nail changes
- enthesitis, dactylitis

Prevalence of axSpA may be greater than RA[1-4]

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